How the spaceplane autopilot works

krpcLab · spaceplane · what the program does and what makes it change what it is doing. State of the code on 2026-10-22 evening (defaults fingerprint `fe76eb15`), kRPC 0.7.0, stock control surfaces.

The autopilot takes a winged vehicle (the target is the Mk3 shuttle `qs_shuttle2` and its family) from orbit to a stop on the KSC runway. It does not follow a stored plan. Every tick it reads the vehicle's state, propagates the rest of the flight to the ground with an RK4 integrator (`trajectory.py`), and takes each decision from that fresh prediction. `autopilot.py` is the phase machine and the only code that talks to the game; `guidance.py` holds the control laws as pure functions; `environment.py` holds the aerodynamic table the vehicle measures for itself; `vacuumown.py` and `allocate.py` hold the client-side attitude arithmetic; every tunable is in `config.py`.

The phase machine

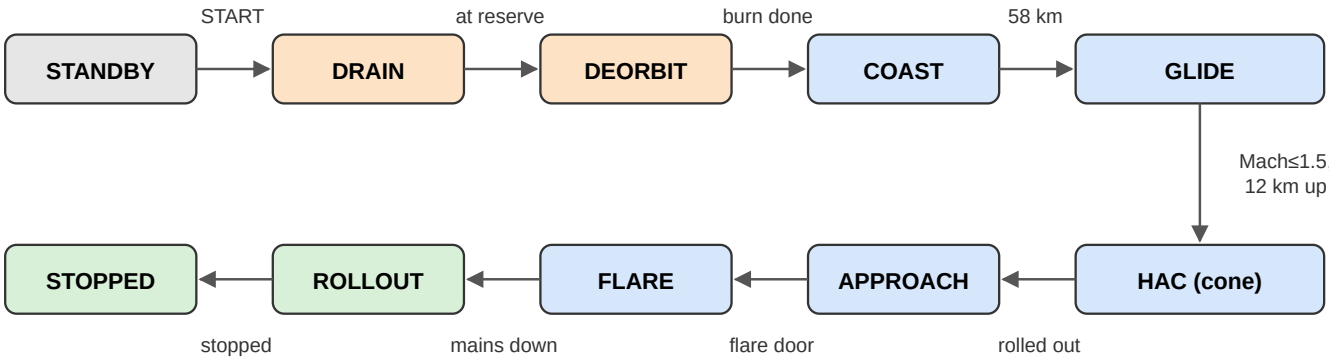


Figure 1. Phases and the conditions that advance them. Orange phases use the engine; blue are unpowered flight; green is on the ground. Engaged below the interface, START skips to GLIDE, APPROACH or FLARE by height and distance to the runway.

Things that run in every phase

mechanism	what it does, and when
Prediction	A propagation to the ground of the law the vehicle will fly (the same function the controller calls), from the current state, each tick. The deorbit and the glide solve against it; the cone and the approach plan from it.
Aerodynamic table	Swept in STANDBY with kRPC's `simulate_aerodynamic_force_at` over angle of attack −8° to 90° and Mach; rows are re-probed in flight. Lift is signed on the rows below zero alpha, and a table with such rows is read signed rather than mirrored. Stall speed (an equivalent airspeed), best L/D and the approach speed are read from it, not configured.
Attitude	kRPC 0.7.0's attitude autopilot flies every phase (the exception is the optional `VACUUM_OWN` coast, below). Its PID is auto-tuned to a per-axis `time_to_peak` derived from the craft; Figure 2 shows how that tune moves through the flight. kRPC's wobble mitigation is off and its pointing deadband stays at its 1°. Roll is engaged at any pointing error (`roll_engage_angle` 175°), because bank is the control.
Bank slew	In GLIDE, HAC and APPROACH the bank command never steps (`guidance.ease_bank`): it moves no faster than the measured roll rate, accelerates and brakes at half the bank acceleration the wheels' and surfaces' torque allows at the alpha flown (RCS left out, so the slew does not depend on the valve), and arrives at the new lean with no rate to carry past it.
RCS valve	Burn and coast: open when the pointing error exceeds 5°, closed 2 s after it is under 1.5°. GLIDE through FLARE: armed, and fires on saturation — once an axis's summed input has sat at ≥ 0.95 for 0.5 s while the pointing error has not closed by 20%; closed once the input has stayed under 0.6 for 2 s. A bank reversal holds it open. Shut above 20 kPa.
Alpha ceiling	Learned: when the vehicle sits well below its commanded alpha, the ceiling ratchets down to what it holds, and the prediction uses the same limit (`trajectory.Holdable`, binned by dynamic pressure).
Pitch trim	In the cone and approach the nose canards are driven as a trim that walks the standing pitch input to zero (`CANARD_TRIM`, alpha under 15°), so the standing moment sits on a surface and not in kRPC's

	integrator. In vacuum the surfaces are held at neutral (`VACUUM_SURFACE_HOLD`).
Ballast	The propellant left after the burn is pumped into each kind's forward-most tank on the coast's first tick (`FUEL_TO_NOSE`), so the centre of mass sits where the airframe trims.
Brakes	Built from the control surfaces and probed in STANDBY: the split rudder (fins opposed: drag with no yaw or roll) and the spoiler (opposed flaps). Used only below the glide, in the order split rudder → spoiler → weave/S-turn.
Time scale	The farm's time-scale plugin is set so each tick's game interval stays what the phase asked for; every log ends with a `loop rate` line.

The attitude tune, phase by phase

Static tune. At engagement each axis gets `ATTITUDE_SLEW_FACTOR` × √(I/τ), with I the axis's moment of inertia and τ the torque available without the air (wheels; yaw counts the RCS while the valve is open). The factor is 1.0. The tune is handed over in kRPC's order: pitch, roll, yaw.

Pitch in the air. From the cone on, pitch is retuned as the surfaces' authority grows with dynamic pressure (`retune_pitch_air`: √(I/(wheels + surfaces)), smoothed over 5 s, floored at 3 s). It is checked every second and re-assigned only past a 15% change, because kRPC re-derives its gains on every assignment.

Roll damper. `rollrate.RollDamper` multiplies roll's time_to_peak by 1.5 for each bank swing that grows by more than 10% about a steady command. It relaxes it back toward the floor over 120 s. Yaw keeps its static figure; while the RCS valve is open in the glide or the cone it flies what wheels plus RCS deliver, never faster than roll and never slower than static (`yaw_time_to_peak`).

Lateral scale in the cone and on final (`APPROACH_SLEW_FACTOR` 1.91, `APPROACH_SLEW_IN_HAC`). In HAC, APPROACH and FLARE, the roll and yaw time_to_peak actually given to kRPC are 1.91 × the tune above; pitch is not scaled. The reason: kRPC aims at each target as an attitude to stop at. At the stiff glide tune its yaw loop holds a slip against a steady bank, which cancels the bank's horizontal lift, so the vehicle banks and does not turn. The looser lateral tune lets the turn happen. The scale is applied on entering the cone.

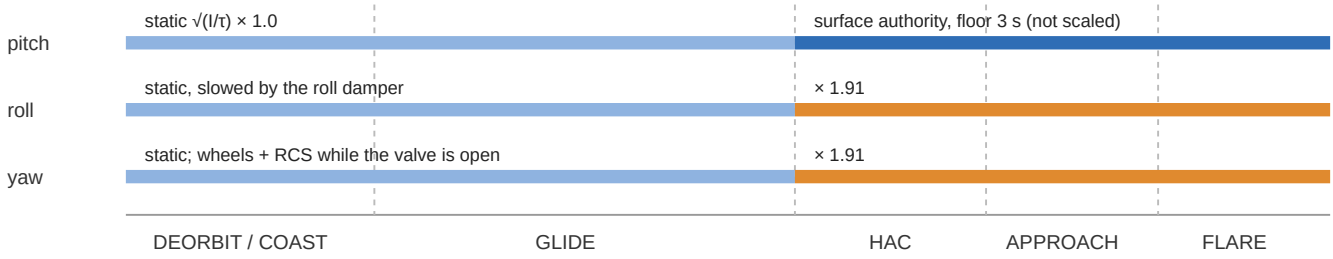


Figure 2. Where each axis's kRPC tune comes from. Light blue: the static derivation and its damper/yaw schedules. Dark blue: pitch follows the surfaces' authority from the cone on. Orange: roll and yaw at 1.91 × their tune from the cone through the flare.

Phase by phase

STANDBY — until START is pressed

Configures the craft (main gear brakes 200% / friction 10, nose gear brake off), measures wheel clearance and tail-strike angle, tunes the attitude controller, sweeps the aerodynamic table, and finds the brakes it can build from the surfaces. On START: below the 58 km interface → GLIDE / APPROACH / FLARE by height and distance; otherwise → DRAIN.

DRAIN — before the burn

Opens the drain valves and empties propellant down to a reserve of 200 m/s of burn (or 1.25 × the burn that puts the periapsis at the surface, if more), at the engine's vacuum Isp. With `VACUUM_OWN` on, the input matrix is measured here, at rest (below). → DEORBIT.

DEORBIT — the latest window, line up, one burn

- **Search** (`deorbit_solution`): a grid, then refinement, over retrograde dv 10–400 m/s as an impulse.
  - For each candidate it propagates the two corners of what the glide may later command (`deorbit_window`): the shortest entry (high drag) and the longest (most lift).

    - The burn chosen puts the gate 25% of the half-width short of the window's centre.
    - Each corner's peak heating index ($\sqrt{\rho}\cdot v^3$) and the window's width are kept with the answer.
- **Plan, first window.** The orbit is propagated forward in vacuum and the search is asked of future states every 60 s (bisected to 8 s), up to three Kerbin days ahead. A state qualifies when all three hold:
    - the runway is within the glide's reach along track (interface to gate at most 1400 km);
    - it is within reach sideways (the gate's offset from the ground track at most half the reach of an entry leaning one way);
    - its burn costs no more than the apoapsis burn that puts the periapsis at the surface.
- **Plan, then the latest** (`DEORBIT_LATEST`, Figure 3). From the first qualifying state the scan keeps stepping forward while a burn still qualifies and:
    - its window is at least 15% as wide as the first window's;
    - its hotter corner's peak heating is no more than the larger of 650 and the first window's own;
    - its dv is no more than a surface periapsis costs.

  The plan is the latest qualifying state at least 60 s inside the last one. On the 80 km orbit that is about 62 m/s, and burn to stop takes about 1100 s.
- **Wait.** A stock maneuver node is placed at the window and `warp_to` takes the game to 60 s before it. A window not seen 120 s after it was due is replanned.
- **Burn lead.** The live trigger asks the search of the state half a burn ahead, so the finite burn straddles the impulsive point.
- **Line up.** The game's SAS holds Maneuver, RCS on. The burn commits once the nose is within 3° on two ticks in a row, or after a hold stuck inside 8° for 6 s, or after 120 s.
- **Burn.** SAS holds the committed attitude, and thrust is limited so the burn lasts at least 3 s. The stop test re-propagates the arc each tick. At cutoff → COAST.

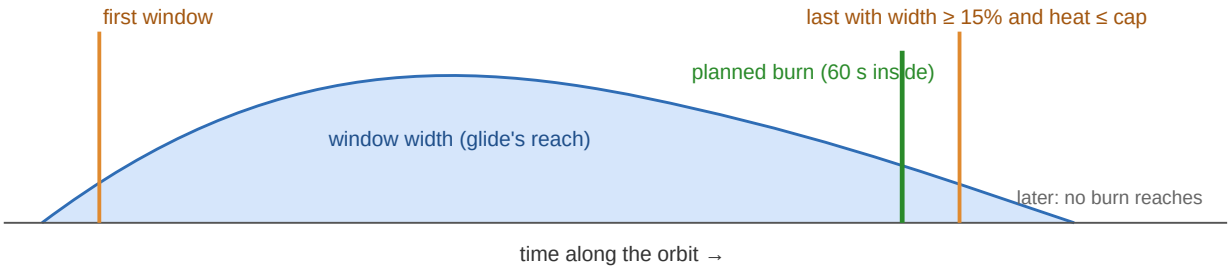


Figure 3. The latest-window rule. Later states are closer and need a larger, steeper burn, which leaves the glide less room; the plan takes the latest state that still leaves 15% of the first window's width and runs no hotter than the cap.

COAST — to the 58 km interface

The coast's first tick pumps the propellant into the nose (`FUEL_TO_NOSE`). `warp_to` then takes the game through the part above the atmosphere; the vehicle keeps the burn's retrograde attitude through the warp. The warp is released 2 km above the air. In the last minute above the air the vehicle swings to the entry attitude, 22° alpha and 30° of bank toward the runway, with the RCS valve permitted (`COAST_RCS_TURN`); a turn of up to ~155° in under a minute is beyond the reaction wheels. The bank's side is latched so the glide starts on the same lean. At 58 km → GLIDE.

GLIDE — the entry, Mach 25 to Mach 1.5

Each tick (1 s) the glide solves alpha and bank against the prediction. Above Mach 4.5 the solve nulls the along-track miss where the arc crosses 12 km. At Mach ≤ 4.5 (latched) it nulls the energy handed to the cone: the predicted energy height $h + v^2/2g$ at the cone's entry, against 12 km at the gate speed plus a 1500 m margin (≈ 14.1 km).

control	role
alpha (20–40°)	Downrange / energy: more alpha, more drag. Rate-limited to 3°/s, held still during a reversal, clamped by speed and by the learned ceiling. In the terminal glide the prediction flies a 10 s mean of the achieved alpha.
bank magnitude (30–70°)	Takes what alpha leaves: rolling lift off vertical sinks the vehicle into denser air.

bank sign	Cross-track: lean toward the runway; reverse when the track's angle to the gate or the predicted cross-track leaves its deadband.

At Mach ≤ 1.5, and either 12 km over the field or within 3 km of the gate, the glide releases the alpha ceiling, picks the runway end and the turn direction (`hac_choose`), and hands over → HAC.

HAC — the heading-alignment cone

- **Radius.** The circle, tangent to the extended centreline and ending at the 2 km gate, whose path spends the energy above the gate (height plus the speed above the gate's target). The path is priced down a ladder of 500 m slices at the L/D the vehicle flies there. Radius 2–16 km, rate-limited; a lap is added only when even the widest circle is short of path.
- **Speed** (Figure 4). The target is the larger of two speeds:
  - *Constant dynamic pressure* (`HAC_SPEED_Q`): the speed whose q equals the gate target's at the height flown. The gate target is the approach speed, 2.25 × the measured stall.
  - *The energy speed* (`HAC_ENERGY_SPEED`): the fastest speed whose own L/D spends 90% of the surplus over the tightest circle holdable at that speed. It is capped where lift at the −5° alpha floor still carries the weight, and by what drag could still bleed to the gate's speed.
- **Alpha holds the speed:** the trim alpha for the banked load plus 0.2° per m/s of excess, between −5° and 22°, capped at 3 g of lift.
- **Brakes:** the split rudder first, opened to the angle whose L/D factor spends the height to the gate; the spoiler second; the weave last (up to 50° off track).
- **Lateral tune:** roll and yaw at 1.91 × (Figure 2).
- **Exit:** lined up, within 900 m of the gate, no lap left, and no more height than the approach can absorb → APPROACH. Below 2 km → APPROACH regardless.

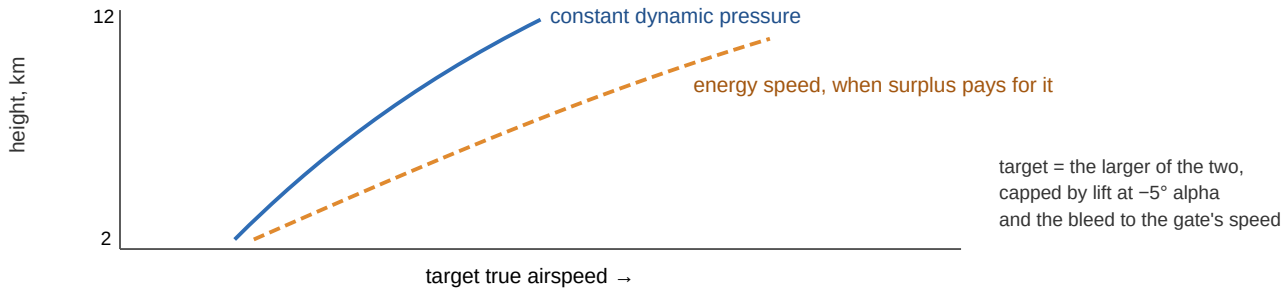


Figure 4. The cone's speed target against height (shape only). On the shuttle the q target is about 112 m/s true at the 2 km gate and 245 m/s at 12 km.

APPROACH — final, 2 km to the flare

- **Aim point:** 1200 m past the threshold (`TOUCHDOWN_AIM_M`). The approach aims 1000 m nearer, so the flare's float lands in the zone; touchdown is about 600–800 m past the threshold.
- **Speed, not path:** alpha is set from the descent angle that holds the target speed (2.25 × stall), from $dv/dt = g \sin\theta - D/m$. The trim comes from the table, including negative alpha; the command is floored at 0°.
- **Brakes:** the split rudder to the L/D factor the geometry asks for; the spoiler when the rudder is at its stop; S-turns when more than 200 m of excess height remains and both brakes are at their stops.
- **Lateral:** a closure that brings cross-track to zero as the wheels arrive, with a 6° bank floor. Roll and yaw stay at 1.91 ×.
- **Divert north** (`ABORT_NORTH`): if the energy height over the distance to the threshold, at the table's best glide ratio, falls short by more than 10% of the height left, the tracked line is latched 150 m north. The space centre lies south.
- **Flare door:** at 50 m + 2.5 s × sink rate → FLARE.

FLARE — the last 50–100 m

Stows the brakes. Tracks a sink rate that eases to the touchdown rate, with alpha capped by maximum lift and by the tail-strike angle. A small lean continues the lateral closure down to the wings-level height; below 30 m the nose is aligned with the runway. Gear goes down. Main wheels touching → ROLLOUT.

ROLLOUT — on the runway

Ground spoiler out on contact; the nose ramped down over 1.5 s; full main-gear braking; nosewheel steering onto the centreline. If the stopping distance runs past the far end, the line steered to is latched 150 m north; it is computed at the larger of the last second's deceleration and the average since touchdown. At the stop speed → STOPPED, which logs the stop and hands the vessel back.

Which law decides what

quantity	DEORBIT	GLIDE	HAC	APPROACH	FLARE / ROLLOUT
when / where	latest window with ≥ 15% of the first's reach, heat ≤ cap, dv ≤ a surface periapsis	—	—	aim 1200 m past the threshold	—
downrange / energy	burn dv	alpha, then bank: range above Mach 4.5, cone-entry energy below	radius and laps; split rudder → spoiler → weave	split rudder → spoiler → S-turns; divert north when short	divert north on an overrun
speed	—	speed cap on alpha	alpha to max(q target, energy speed), −5° floor	descent angle that holds speed	sink-rate law / wheel brakes
cross-track	sideways reach test	bank sign deadbands	circle geometry	closure, bank floor 6°	bank ≤ 12°, nosewheel
attitude	stock SAS	kRPC 0.7 attitude autopilot, tuned from the vessel; roll/yaw × 1.91 from the cone on (Figure 2)			

Optional modes (off by default)

These are in the code behind flags, for craft the defaults do not fly. Each acts only when its flag is set with `--set`.

`VACUUM_OWN` **— the coast flown on a measured input matrix.** This is for a craft kRPC cannot point in vacuum: one whose RCS inputs cross-couple, so that a yaw input also makes pitch or a roll input mostly yaw. kRPC assumes each input turns its own axis, so its corrections land on the wrong axes.

1. **Measure (DRAIN or DEORBIT, at rest in orbit).** kRPC is let go and each input is pulsed +1 then −1 for 1 s. The change in body angular velocity over each pulse gives one column of B, the angular acceleration per unit input. The +/− difference cancels any drift. From B the largest pure acceleration each axis can be given is computed.
2. **Turn before the warp (COAST's first ticks).** The arc is propagated to the atmosphere's edge, and the entry attitude is computed for the velocity there: the coast's alpha above it, rolled to its bank. The client flies to it, and the warp is held until the vehicle is within 3° and turning at under 0.3°/s, or for at most 300 s. Rails warp keeps the attitude.
3. **After the warp.** The coast's own commanded attitude is flown, which corrects the planet's rotation during the warp.
4. **Hand back.** At 150 Pa, kRPC flies again.

The steering law (`vacuumown.command`), each tick at 0.1 s:

- The error is the rotation taking the body axes onto the target's.
- Per axis, the rate to turn at is the bang-bang approach to that error on half the axis's authority, capped at 0.2 rad/s.
- The acceleration asked is the rate error over 1 s, and the inputs are B^{-1} times it. When any input would exceed 1, all three are scaled down together, so the direction of the correction survives.
- **Nose first:** while the nose is more than 15° off, the roof is held where it is. Roll is usually the weak axis, and asked together it would starve the pitch.

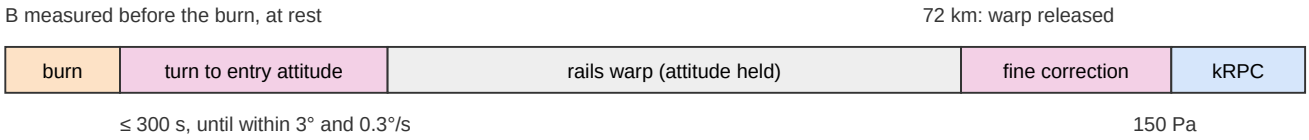


Figure 5. The coast under `VACUUM_OWN`. Pink: the client flies B^{-1}. Grey: warp. Blue: kRPC from 150 Pa on.

flag	what it does
`ATTITUDE_PITCH_AIR_GLIDE`	Pitch's air-authority retune (see Figure 2) in the GLIDE as well, not only from the cone on. It is for an airframe with little pitch stability of its own, which the static (wheel-sized) pitch loop cannot hold hypersonically.
`COAST_ROLL_FREE_Q_PA`	Below this dynamic pressure the coast and glide tilt the nose alpha above the wind toward the roof the vehicle already has, and ask no roll; the commanded bank is flown above it.
`SURFACE_OWN`	The client flies the attitude from GLIDE to FLARE. kRPC is disengaged and every surface is overridden. The surfaces are allocated by their measured moments (swept in vacuum before the burn), with the RCS as three more actuators, by incremental dynamic inversion.

Logs

Nothing prints. Each run writes `logs/LOG<n>`:

- a `config:` line: the fields that differ from the defaults, plus the defaults fingerprint;
- one telemetry line per tick interval;
- every phase change and decision as an event;
- a closing `loop rate` line.

Telemetry fields worth knowing:

field	meaning
`bnk=`	the bank the roof is at
`fbk=`	the bank and sideways g of the *aero force* about the wind. A roof banked with the force not banked is side force (slip, rudder) cancelling the turn.
`yerr=`	the yaw and roll errors kRPC is handed: the commanded nose toward the body's right, and the commanded roof about the nose
`kerr=`	kRPC's own pitch/heading/roll errors, from the cone on
`yrin=`	the yaw and roll inputs the vessel receives
`vt=`, `ves=`	on the cone's line: the speed alpha flies to, and the energy speed
`vown=`, `u=`	under `VACUUM_OWN`: the attitude error per body axis, and the inputs sent

Events worth knowing:

- `deorbit latest:` the span of qualifying burns, the plan, the dv and the heating;
- `attitude time_to_peak x1.91 in HAC`: the lateral scale being applied;
- `vacuum own:` the matrix B, the entry attitude, whether it was held before the warp, and the hand-back.

The `DOWN:` line gives the stop along and across the runway and whether it is on the tarmac: the measure of a landing.